

REVIEW ARTICLE

General anaesthesia for nonobstetric surgery during pregnancy

A narrative review

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Nonobstetric surgery may be required in up to 1% of pregnancies. Most common procedures are urgent abdominal surgeries requiring general anaesthesia. Maintaining normal maternal physiology during anaesthesia is vital. Left lateral tilt position prevents aortocaval compression and preserves normal venous return. Preparation for a difficult airway is required in all obstetric patients due to the known anatomical and physiological changes. Aspiration prophylaxis and a rapid sequence induction are traditionally recommended to avoid the (probably overestimated) risk of aspiration. Although the minimum alveolar concentration (MAC) of volatile anaesthetic agents is reduced by 30 to 40%, awareness occurs more frequently in the obstetric population. Maternal outcomes from surgery are comparable with those of nonpregnant women, but increased incidences of foetal loss, preterm delivery, low birth weight and caesarean section have been reported. Although animal studies have

observed impaired foetal brain development after antenatal exposure to anaesthesia, the translational value of these studies remain controversial. Clinical evidence is nearly absent. Withholding urgent/essential procedures is certainly more threatening than proceeding with the surgery. To increase the safety of mother and foetus, nonurgent or nonessential procedures should be postponed until after delivery, and if procedures cannot wait, locoregional anaesthesia should be used if possible. Where general anaesthesia cannot be avoided, the duration of exposure should be kept to a minimum. This narrative review summarises the literature of the past 20 years concerning the anaesthetic management and outcomes of nonobstetric surgery under general anaesthesia during pregnancy.

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KEY POINTS

- Nonobstetric surgery is required in up to 1% of pregnant women.
- Maintenance of normal maternal physiology during anaesthesia is vital.
- Maternal outcomes from surgery are comparable with those of nonpregnant women, but increased risks for foetal loss, preterm delivery, low birth weight and caesarean section have been reported.
- Although animal studies have observed impaired foetal brain development after antenatal exposure to anaesthesia, their translational value remains controversial.

- Urgent and essential procedures must not be postponed during pregnancy.

Introduction

Nonobstetric surgery under anaesthesia is required in 0.39 to 0.96% of pregnant women.^{1–4} Abdominal procedures (35 to 81%) are most commonly indicated, followed by trauma (8 to 24%) and urological surgery (8 to 16%).^{1–6} Appendicitis (6 to 39%) and adnexal diseases (6 to 36%) are the most common diseases requiring surgery during pregnancy,^{1,3,4,7–9} most (68 to 91%) being emergencies.^{4,6,9,10}

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In general, the incidence of surgical procedures is comparable in pregnant and in nonpregnant women.⁷ Exceptions are cholecystectomies, adnexal disease, cardiac surgery and lesions in the central nervous system, which are more frequent during pregnancy than outside pregnancy.^{7,11–13} Appendicitis has more severe postoperative complications (e.g. peritonitis and sepsis) in pregnant women when compared with nonpregnant patients due to diagnostic difficulties and delayed treatment.^{13–16}

Careful consideration of pregnancy-induced physiological changes is required in order to provide well tolerated and effective anaesthesia to the pregnant patient. The aim of this narrative review is to provide an update on the management of general anaesthesia for nonobstetric surgery during pregnancy and will include discussion regarding maternal and foetal outcome.

Materials and methods

PubMed and Embase were searched from January 1, 2001, until May 18, 2021, with a search string comprising the concepts anaesthesia, pregnancy and surgery (supplement 1, <http://links.lww.com/EJAIC/A0>). As a first step, all records were systematically screened for eligibility. Inclusion criteria were original studies and reviews reporting on the anaesthetic management for general anaesthesia during pregnancy for nonobstetric surgery including outcomes. Exclusion criteria were articles discussing general/neuraxial anaesthesia for caesarean section, in-vitro studies, case reports, non-English articles and articles without full text. In the second step, the

reference lists of included articles were further screened for relevant articles. These publications did not meet all eligibility criteria if no other publications on a relevant subject were available. Two authors (T.B. and S.D.) obtained a consensus for inclusion of articles.

Results

In PubMed, 3198 and in Embase, 5701 records were found. After removal of duplicates, 7218 records remained of which 49 met the eligibility criteria of the first step. After consulting the reference lists of these articles in a second step, 86 additional publications were identified. For example, articles older than 20 years were used when more recent publications were not available. Another example is that no original studies could be found on the choice of vasopressors during general anaesthesia for nonobstetric surgery, but reviews meeting the eligibility criteria were found referring to studies of neuraxial anaesthesia for caesarean section.

Anaesthetic management

A summary of the recommendations is provided in Table 1.

Aspiration

In 1946, Mendelson¹⁷ reported that pregnant women *in labour* were at risk for aspiration during anaesthesia. This publication formed the basis for the dogma that gastric emptying in pregnant patients is delayed. Moreover, 40 years ago, one study¹⁸ demonstrated that the intragastric pressure is increased from early pregnancy,

Table 1 Summary of practical recommendations

Practical recommendations/study data	
Aspiration	● Fasting 6 to 8 h for food and 2 h for clear liquids. ● Gastric emptying is nearly normal during the entire pregnancy until the onset of labour. ● Expert opinion recommends rapid-sequence induction and aspiration prophylaxis from 15 to 18 weeks of gestation. ● Gastric ultrasound is a reliable tool to evaluate the residual stomach content. ● Extubation: fully awake patient and train of four ratio is >90%.
Airway and ventilation	● Rapid desaturation during apnoea. ● Pre-oxygenation until end-tidal oxygen concentration $\geq 90\%$. ● Gentle bag-mask ventilation (<20 cmH₂O) is recommended in combination with cricoid pressure during the rapid sequence induction. ● Difficult airway and failed intubation is more frequent than in nonpregnant women. ● Maintain the physiological $P_a\text{CO}_2$ of 26 to 32 mmHg.
Haemodynamics	● The blood flow to the pregnant uterus is pressure dependent without autoregulation. ● Near-normal maternal blood pressures must be maintained. Targets stated in literature are 70, 80, 90% of baseline or a mean arterial pressure >65 mmHg. ● Left lateral tilt ($\geq 15^\circ$ to 30°) position needs to be used starting from 10 to 18 weeks of gestation.
Transfusion, coagulation, thromboprophylaxis	● A haemoglobin concentration less than 11 g dl^{-1} in the first trimester and $<10.5\text{ g dl}^{-1}$ in the second and third trimester are defined as anaemia during pregnancy. ● A maternal haemoglobin level $<6\text{ g dl}^{-1}$ is associated with impaired foetal outcome, but there are no clear criteria to initiate a red blood cell transfusion. ● Transfused red blood cells should be cytomegalovirus negative; ABO, rhesus and Kell compatible ● Fibrinogen concentration is higher during pregnancy. ● Pharmacological and nonpharmacological thromboprophylaxis must be discussed after each surgery.
Awareness Monitoring	● There is an increased risk for awareness in pregnant women undergoing nonobstetric surgery. ● Mother: standard American Society of Anaesthesiologists monitoring,^{23,50,52,61} depth of anaesthesia and quantitative train of four monitoring before extubation. ● Foetal heart rate monitoring can be used.
Tocolysis	● Prophylactic tocolysis is not recommended. ● Tocometry is recommended in the early postoperative phase to allow start of tocolysis if necessary.

Table 2 Aspiration prophylaxis

Drug	Time of administration
Metoclopramide 10 mg intravenous	30 min before induction
Oral sodium citrate 0.3 mol l ⁻¹ 30 ml	15 min before induction
Ranitidine	• Elective procedure: 150 mg oral on the evening before the procedure and 2 h before the procedure • Emergent surgery: 50 mg intravenous as soon as possible

whereas the tone of the lower oesophageal sphincter is decreased.^{7,12,18–30} However, it was demonstrated in 1991 that gastric emptying is nearly normal *during the entire pregnancy*^{31–33} *until the onset of labour*.^{7,20,21,24,26,28,29,32,34–36} Recent studies showed that parturients undergoing both elective and urgent caesarean sections being ventilated with a laryngeal mask had no increased risk of aspiration.^{37–48} In most of these studies, the laryngeal mask was inserted after a rapid-sequence induction. In three studies,^{37,41,42} bag-mask ventilation was performed (< 1 min^{37,41}) with⁴¹ or without^{37,42} cricoid pressure. Of note, in all mentioned studies, obese patients and women suffering from gastro-oesophageal reflux were excluded.

Notwithstanding the most recent data, expert opinion still recommends performing a rapid-sequence induction^{12,20,21,24,26,29,30,35,49,50} and to administer aspiration prophylaxis (Table 2), starting from 15 to 18 weeks of gestation.^{7,12,19,21,23,25,26,35,50–52}

Similar to nonpregnant patients, fasting times of 6 to 8 h for food and 2 h for clear liquids need to be respected.^{12,21,35,50,52} Gastric ultrasound is a reliable tool for evaluating the residual stomach content.³⁶ Extubation

is only allowed when the patient is fully awake and the train of four ratio is more than 90%.^{7,21,23,29,50,52}

Airway and ventilation

During pregnancy, functional residual capacity (FRC) is decreased and oxygen consumption is increased (Table 3), resulting in rapid desaturation during apnoea.^{7,12,13,19,20,22,23,25–27,29,30,51,53,54} Pre-oxygenation with 100% oxygen must be performed before induction until end-tidal oxygen concentrations of 90% are achieved.^{7,12,22,24,25,52} After 3-min tidal volume breathing or eight deep breaths, this target was achieved in only 76% of pregnant women during their third trimester.⁵⁵

After induction of anaesthesia, gentle bag-mask ventilation (<20 cmH₂O) is recommended in combination with cricoid pressure during rapid-sequence induction, as it can decrease oxygen desaturation and allows estimation of the difficulty of bag-mask ventilation, which could be necessary during prolonged or failed intubation attempts.⁵⁶ If bag-mask ventilation is difficult, an oropharyngeal airway, a four-handed (two-person) technique, should be used and cricoid pressure should be released.⁵⁶

Table 3 Cardiovascular and respiratory changes during pregnancy

	Parameter	Change
Cardiovascular system	Cardiac output	↑ 40 to 50% ^{13,19,25,28,30}
	Stroke volume	↑ 20 to 30% ^{7,28}
	Systemic vascular resistance	↓ ^{13,19,22,25}
	Pulmonary vascular resistance	↓ ^{22,25}
	Plasma volume	↑ 40 to 50% ^{13,25,28}
	Red blood cell mass	↑ 20 to 30% ^{13,25}
	Haematocrit	↓ 20% ²⁴
Respiratory system	Functional residual capacity	-↓ 20% in standing position ^{7,20,23–25,28,30} -↓ 50 to 70% in recumbent position ^{22,30}
	Residual volume	↓ 15% ^{25,28}
	Vital capacity	= ²⁸
	Total lung capacity	↓ 5% ²⁸
	Tidal volume	↑ 35 to 45% ^{7,22,28,30}
	Respiratory rate	= ^{22,28} / ↑ 15% ⁷
	Minute ventilation	↑ 35 to 70% ^{7,19,22,25,30}
	Oxygen consumption	↑ 20 to 60% ^{12,13,20,23–25}
	pH	7.40 to 7.46 ^{7,19,22,59}
	P _a CO ₂	26 to 32 mmHg ^{7,19,22,23,59}
	P _a O ₂	95 to 100 mmHg ⁵⁹
	HCO ₃ ⁻	18 to 22 mmol l ⁻¹ ^{122,59}

Increase with x% (↑ x%), decrease with x% (↓ x%) or unchanged (=) when compared with nonpregnant women. Partial arterial carbon dioxide tension (P_aCO₂), partial arterial oxygen tension (P_aO₂) and bicarbonate (HCO₃⁻).^{7,19,22,23,54,59}

Intubation is more difficult during pregnancy because of enlarged breasts, mucosal oedema and more friable mucosa.^{7,12,13,19,20,22–26,28,30,49,51,53,54} Mallampatti scores are higher in pregnancy, and the risk of failed intubation (1/224) is eight times higher than the nonpregnant population.^{7,11–13,22,24,26–28,30,51,54} The patient should be positioned in a reverse Trendelenburg to improve laryngeal visualisation during intubation, decrease passive gastro-oesophageal reflux, optimise breast position and increase FRC.^{23,25,30,52} Preparation for a difficult airway including knowledge of airway management algorithms^{56–58} is essential.

In the healthy pregnant patient, increased minute ventilation results in a lower physiological $P_a\text{CO}_2$, while the $P_a\text{O}_2$ remains unchanged or slightly increased (Table 3).^{12,13,19,24,27,28,30,53,54,59,60} It is recommended to maintain these physiologic values during anaesthesia: high levels of $P_a\text{CO}_2$ could result in foetal acidosis, whereas low values cause uterine vasoconstriction.^{7,13,19–25,53,54,61} Both hypoxia and hyperoxia should be avoided.^{7,19,20,25,52,62}

Haemodynamics

Cardiac output starts to increase in the first trimester and reaches its maximum in the second trimester.^{7,12,13,24,27–30,51} Maternal blood pressure decreases during the first trimester, reaches its nadir in the second trimester and increases again in the third trimester^{7,27,63} (Fig. 1).

The blood flow to the pregnant uterus is pressure dependent without autoregulation; consequently, near to normal maternal blood pressures must be maintained,^{7,12,19–21,23,24,27,29,50,51,54,60} as a drop in blood pressure will directly result in decreased uteroplacental perfusion. Furthermore, it is a general consensus in

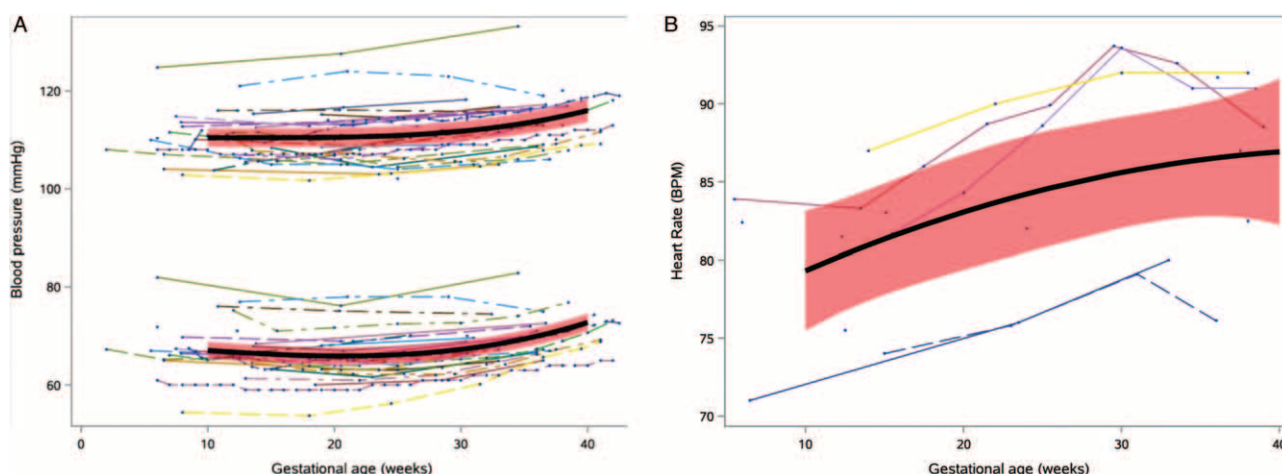
nonpregnant adults that intra-operative hypotension should be avoided, as it is associated with peri-operative acute kidney injury, myocardial injury and increased mortality.⁶⁴ Although most articles recommend to ‘maintain normotension’ or to ‘treat hypotension’ in pregnant women, they fail to define ‘normal’ blood pressure. In contrast, four publications were identified advising specific target blood pressures in pregnant patients: 70%,⁶⁵ 80%,⁶¹ 90%⁴⁹ of baseline or a mean arterial pressure more than 65 mmHg.⁶⁶ Although ‘normal’ blood pressure is only weakly defined, maternal blood pressure should be maintained using vasopressors (see below), intravenous fluids, left lateral tilt position and by avoiding an inappropriate depth of anaesthesia. Nevertheless, there are no well conducted clinical studies investigating the effects of treatment of hypotension during general anaesthesia in pregnancy on the outcome of the mother or foetus.

Starting from a gestational age of 10 to 18 weeks, it is advised to position pregnant women in left lateral tilt ($\geq 15^\circ$ to 30°) to achieve leftward displacement of the uterus, thereby avoiding aortocaval compression.^{7,12,13,19,21–25,27,28,30,35,49–51,53,61,65} However, this theory has recently been challenged: most patients compensate in supine position by increasing blood flow through the azygos system and maintain normal blood pressures.⁶⁷

Transfusion, coagulation and thromboprophylaxis

At 32 weeks of gestation, plasma volume is increased by 40 to 50%, while the red blood cell mass is increased by only 20 to 30%,¹³ resulting in dilutional anaemia.^{7,28,30} Anaemia during pregnancy is defined as a haemoglobin concentration less than 11 g dl^{-1} in the first trimester and less than 10.5 g dl^{-1} in the second and third trimester.^{30,68,69} A maternal haemoglobin level less than 6 g dl^{-1}

FIGURE 1 Reference values for blood pressure and heart rate of awake pregnant women.



This graph of a meta-analysis⁶³ is reprinted with permission. (a) SBP and DBP. (b) Heart rate. The solid black line represents the average, and the red band displays the 95% confidence interval. The thin dotted lines are the data of individual studies included in the meta-analysis.

is associated with impaired foetal oxygenation, abnormal foetal heart rate (FHR) patterns and foetal death.^{30,70} There are no clear criteria to initiate a red blood cell transfusion. This should be individually based on the history, clinical signs (e.g. ongoing bleeding), physiological transfusion triggers and the level of haemoglobin.^{68,71} Only red blood cell units from cytomegalovirus seronegative donors should be used and the transfused red blood cells should be ABO, rhesus and Kell compatible.^{68,71} In case no cross-matched donor blood is available, O-negative blood should be used in rhesus-negative women.^{68,72,73} In case of (abdominal⁷⁴) trauma, bleeding or foetal death, anti-D immunoglobulin should be given to rhesus D-negative women and Kleihauer-Betke test (quantification of foetal haemoglobin) should determine the need for additional doses.^{72,74}

Fibrinogen concentration is increased during pregnancy: the normal ranges during the first, second and third trimesters are 2.4 to 5.1, 2.9 to 5.4 and 3.7 to 6.2 g l⁻¹, respectively (nonpregnant adults: 2.3 to 5.0 g l⁻¹).^{28,75} In the setting of postpartum haemorrhage, early administration of fibrinogen concentrate is preferred over the administration of fresh frozen plasma.^{76–79} Thromboelastometry tests (e.g. ROTEM) allow the bedside estimation of fibrinogen levels.⁸⁰ Fibrinogen substitution is indicated for postpartum haemorrhage when the ROTEM - FIBTEMA5 value is less than 12 mm, corresponding with plasma fibrinogen concentrations less than 2 g l⁻¹.^{81–83} However, the relevance of these thresholds in the setting of nonobstetric surgery is not known.

The risk of thromboembolic complications is increased both by pregnancy (five times higher risk) and surgery. Early ambulation, pharmacological prophylaxis with low-molecular weight heparins and mechanical thromboprophylaxis (compression stockings) are essential.^{7,13,19,21,23,25,30,50,52,53,65}

Awareness

Endorphins and progesterone increase the sensitivity of pregnant women to volatile and intravenous anaesthetics. Therefore, minimal alveolar concentration (MAC) is reduced with 30 to 40% from 8 weeks of gestation.^{7,19,22,28,35,51,52} The incidence of awareness is higher in pregnant women (1/670) than in the general population (1/19600), due to several risk factors (rapid sequence induction, neuromuscular block, difficult airway, high body weight and emergency surgery).^{11,26,84}

Monitoring

Mother

Routine monitoring for general anaesthesia during pregnancy consists of standard American Society of Anesthesiologists (ASA) monitoring,^{23,50,52,61} depth of anaesthesia⁶¹ and quantitative train of four monitoring before extubation.^{50,52} Additional invasive monitoring may be necessary depending on patient comorbidities

and the procedure.^{23,52} End-tidal CO₂ monitoring is also reliable during laparoscopy in pregnancy.^{7,8,21,61,85}

Foetus

Starting from 16 to 22 weeks of gestation, continuous FHR monitoring is technically feasible. FHR variability can be assessed from 25 to 27 weeks of gestation.^{7,13,19–23,25,35,60,61,86} All anaesthetics and opioids easily cross the placenta. This results in a reduced or absent FHR variability and lower baseline FHR.^{7,20–22,25,35,53,60,61,66,87} Suboptimal FHR tracing can indicate hypoperfusion of the uterus and should be managed promptly: maternal homeostasis needs to be restored,^{7,22,35,50,52,88} the pneumoperitoneum or manual retraction should temporarily be released,^{7,52} and the administration of tocolytics should be considered.⁵² Foetal bradycardia can also be an indicator of hypoxia, acidosis, hypothermia or the administration of certain drugs (e.g. neostigmine, opioids).^{7,20} An obstetrician qualified in interpretation of the FHR tracing in the setting of maternal anaesthesia should be present in the operating room and be prepared to perform an urgent caesarean section from 24 to 26 weeks of gestation.^{5,11,21–23,30,35,52,66,86,88} Foetal lung maturation before surgery must be discussed depending on the term of the pregnancy and the expected risk of the procedure.⁸⁹ However, there are no strict criteria as to when a caesarean section should be started in the setting of general anaesthesia (note that reduced or absent FHR variability is normal during general anaesthesia).^{5,11,21–23,30,35,52,66,86,88} Moreover, FHR monitoring and caesarean section can be challenging during certain surgeries (e.g. spine surgery in prone position).^{7,86} Furthermore, there is no evidence that the use of FHR monitoring improves foetal outcome.^{19,35,88} Therefore, the decision to use FHR monitoring needs to be taken both individually and as part of a multidisciplinary approach.^{7,13,86}

Tocolysis

Routine administration of prophylactic tocolytics is not recommended due to adverse effects and lack of evidence of their effectiveness. Tocometry is always recommended in the early postoperative phase to detect premature labour, and to allow treatment to start if necessary, as postoperative analgesia could mask mild contractions.^{7,8,12,19,21–23,25,35,52,61,65}

Drug choices

A summary of the recommendations is provided in Table 4.

Anaesthetics and opioids

Both volatile and intravenous anaesthetics can be used safely.^{24,25,52} Unlike propofol, volatile anaesthetics are clinically used to achieve profound uterine muscle relaxation (e.g. in open or foetoscopic foetal surgery).^{21,24,51–53} As ketamine can transiently increase uterine tone, it should be avoided.²³ Fentanyl, sufentanil, alfentanil

Table 4 Summary of drug choices

	Practical recommendations/study data
Anaesthetics and opioids	• Safe: Volatile and intravenous anaesthetics. • Safe: Fentanyl, sufentanil, alfentanil and remifentanyl.
Neuromuscular blocking drugs	• Safe: All neuromuscular blocking drugs. • Desaturation occurs later after rocuronium than after succinylcholine. • Sugammadex: Placental transfer is probably very limited, but clinical data are absent. • Neostigmine: Can result in foetal bradycardia. Atropine rather than glycopyrrolate needs to be co-administered.
Antibiotics	• Safe: Beta-lactams (e.g. cefazoline, amoxicillin-clavulanic acid), metronidazole, clindamycin and vancomycin • Teratogenic: Fluoroquinolones, tetracyclines and aminoglycosides
Vasopressors and inotropes	• Data are not available for general anaesthesia for nonobstetric surgery. • Neuraxial anaesthesia for caesarean section: ○ Phenylephrine is first choice ○ Ephedrine slightly decreases foetal pH ○ Noradrenaline can be an alternative for phenylephrine
Postoperative analgesia	• Safe: ○ Locoregional anaesthesia ○ Acetaminophen/paracetamol ○ Metamizole in first and second trimester ○ Short period of weak (tramadol) or strong (morphine) opioids • Avoid: ○ NSAIDs ○ Metamizole in third trimester ○ Prolonged use of opioids ○ Ketamine
Postoperative nausea and vomiting	• Safe: Ondansetron, metoclopramide

and remifentanyl can be used safely.^{52,61} All these drugs easily cross the placenta.^{12,20,49-51}

Neuromuscular blocking drugs

Neuromuscular blocking drugs are large molecules that cannot cross the placenta. Their use is considered well tolerated during pregnancy.^{12,23,25,26,35,50,52,53} The use of rocuronium is preferable over succinylcholine, as desaturation occurs earlier after use of succinylcholine.⁹⁰ Moreover, urgent antagonisation of rocuronium by sugammadex can be life-saving. The placental transfer of sugammadex remains largely unknown, but is probably very limited.⁹¹ Sugammadex can encapsulate progesterone *in vitro*,⁹² and unpublished data in rabbits showed a decreased birth weight and ossification impairments after daily administration of supratherapeutic doses.⁹³ In rats, neither decreased progesterone concentrations nor increased abortions were observed, but clinical data are not available.^{91,94} Neostigmine crosses the placenta and can result in foetal bradycardia. Atropine rather than glycopyrrolate needs to be co-administered with neostigmine, because atropine crosses the placenta as opposed to glycopyrrolate.^{12,21,23,25,50,51,91}

Antibiotics

Beta-lactams (e.g. cefazoline, amoxicillin-clavulanic acid), metronidazole, clindamycin and vancomycin can all be used safely.⁹⁵ Fluoroquinolones, tetracyclines and aminoglycosides are teratogenic and should be avoided.⁹⁵

Vasopressors and inotropes

For decades, ephedrine has been the gold standard to treat hypotension during neuraxial anaesthesia for caesarean section. It was found not to decrease uterine blood

flow in sheep^{96,97} in contrast to phenylephrine^{98,99} and (nor)adrenaline.⁹⁸⁻¹⁰² Also, in nonhuman primates¹⁰³ and humans,^{104,105} phenylephrine decreased uterine blood flow, in contrast to ephedrine. In 2002, a meta-analysis of clinical studies examining neuraxial anaesthesia for caesarean section found the foetal pH to be significantly higher [mean difference: 0.03 (95% confidence interval: 0.02 to 0.04)] after the use of phenylephrine compared with ephedrine, but no differences in Apgar scores were observed.¹⁰⁶ In 2015, Ngan Kee *et al.*¹⁰⁷ showed that the foetal outcome after the use of noradrenaline was comparable with phenylephrine. Noradrenaline has some advantages over phenylephrine because it results less frequently in maternal bradycardia, preserves maternal cardiac output and results in lower levels of catecholamines in the fetus.¹⁰⁷

Although phenylephrine is currently still the vasopressor of choice for spinal-induced hypotension for caesarean delivery,¹⁰⁸ there is no consensus on the best vasopressor to treat hypotension during general anaesthesia for nonobstetric surgery.

Postoperative analgesia

Postoperative analgesia is important, as pain can result in stress, reduced uterine blood flow and premature contractions.^{7,21,22,51,53} The following considerations should be addressed individually:

- (1) Use locoregional anaesthesia whenever possible (see below).^{21,22,52}
- (2) Acetaminophen/Paracetamol is the first choice to treat mild to moderate pain.^{13,21,22,52,65}
- (3) NSAIDs should be avoided due to the risk of miscarriage (first trimester), oligohydramnios and

premature closure of the ductus arteriosus (third trimester).^{13,21–23,25,26,52,65}

- (4) Metamizole should be avoided in the third trimester.¹⁰⁹
- (5) Weak opioids (e.g. tramadol) or strong opioids (e.g. morphine) can safely be used for a short period.^{21–23,52,65}
- (6) Ketamine should be avoided, as it may impair foetal brain development.¹¹⁰

Postoperative nausea and vomiting

The peri-operative use of ondansetron and metoclopramide is well tolerated during pregnancy.^{111,112}

Outcome

Maternal safety

In retrospective cohort studies, no significant differences were shown for the incidence of postoperative complications within 30 days after surgical procedures in pregnant versus nonpregnant women.^{3,60,113}

Pregnancy outcomes

Adverse pregnancy outcomes after nonobstetric surgery were recently investigated in retrospective studies.^{1–6,9} The risk of premature (<37 weeks) labour was significantly increased,^{1–3} particularly in patients in the third trimester having lower abdominal surgery.^{1–4,25,60,61} Higher incidences of lower birth weight were also observed.^{1,2,6} Although significantly increased risk for caesarean section was reported in two studies,^{2,3} one study¹ found no significant difference. Miscarriages,^{2,3} pregnancy terminations³ and stillbirths² were more frequent after surgery. The overall risk of adverse pregnancy outcomes was increased by 2% for every 1-min anaesthesia time.⁴

In the last decade, there has been an increase in the use of laparoscopic techniques during pregnancy. There are no differences in obstetric and foetal outcomes (foetal loss rates, preterm labour, caesarean section rates, birth weight and Apgar scores) for laparoscopy when compared with laparotomy.^{8,14,60,65,114} Advantages of laparoscopy include less postoperative pain, earlier return of bowel function and oral intake, shorter recovery times and a lower risk of thromboembolic events.^{7,8,11,13,14,52,65,114} Moreover, pneumoperitoneum may be safer than manual retraction of the uterus during open surgery.^{13,15,52} Laparoscopic surgery is well tolerated and effective and should be considered as the standard treatment.^{1,4,8,9,11,13–15,30,65,114,115}

Teratogenicity

There is no evidence of teratogenicity after clinical use of intravenous anaesthetics, volatile anaesthetics, neuromuscular blocking drugs, local anaesthetics or intraoperative and postoperative use of opioids.^{7,25,26,28–30,52,61,86,116} Nitrous oxide impairs DNA and RNA synthesis,¹¹⁷ but was only teratogenic in rats after exposures

to concentrations of at least 50% for at least 24 h,¹¹⁸ and these effects were absent in humans.^{7,25,26,52,61,119} Some older clinical studies reported congenital malformations associated with benzodiazepines,¹²⁰ but these observations were not reproduced in other studies.^{26,61}

Neurotoxicity

In 1999 and 2003, Ikonomidou and Jevtovic-Todorovic exposed foetal and neonatal rats to general anaesthesia and observed widespread apoptosis in the brain resulting in persistent impairments of learning and memory.^{121,122} In the following years, an increasing number of studies^{123–131} demonstrated in multiple animal species that exposure to commonly used general anaesthetics resulted in neuronal injury and neurobehavioural impairments. Consequently, in 2016, the Food and Drug Administration (FDA) raised a warning that repeated or prolonged exposure of pregnant women during their third trimester to general anaesthesia could impair foetal brain development.¹³²

A recent meta-analysis of preclinical studies confirmed that exposure of pregnant animals to general anaesthesia impaired learning and memory and resulted in neuronal injury in the offspring.¹¹⁰ This was concluded for all investigated animal species, anaesthetic drugs and pregnancy trimesters. However, several methodological issues were identified that limit the translational value of these results to the clinical setting.

Recently, the first retrospective clinical study¹³³ has been published, which investigated the effects of exposure of pregnant women to general anaesthesia. Only one significant difference was observed in neuropsychiatric outcome of the children: more externalising behavioural problems in exposed versus unexposed children. Limitations included an uncertain exposure status, the exposure occurring 30 years ago and a small sample size (22 exposed children).

Discussion

Principal findings

Exposure of pregnant women to general anaesthesia is not uncommon and often unavoidable. Key points to remember during anaesthesia are normal maternal physiology needs to be maintained, awareness occurs more frequently, desaturation occurs faster, the airway can be hazardous and aspiration should be prevented. Maternal outcomes are comparable with those of nonpregnant women; however, adverse pregnancy outcomes are more common. Animal studies have reported impaired foetal brain development, but, to date, there are no clinical data to confirm this finding in humans.

Practical implications

Although nonobstetric surgery during pregnancy is associated with increased obstetric risks, it is not known whether these are attributable to the treatment, to the disease itself or anaesthesia.²⁶ Postponing urgent surgical procedures during pregnancy and choosing medical

treatment instead results in more complications. For example, foetal loss following an uncomplicated appendectomy occurs in less than 1.5% of cases, whereas foetal loss rate after rupture of the appendix is 20 to 35%¹⁵. The maternal odds of septic shock is six times higher following conservative treatment of appendicitis versus surgical appendectomy.¹⁶ Likewise, foetal loss rate is 4% after uncomplicated cholecystectomy versus 10% after cholangitis and 60% after biliary pancreatitis.^{13,14} Hence, urgent and essential procedures must be performed promptly,^{11,13,16,22,30,35,52,60,86} while nonessential or nonurgent procedures should be postponed until after delivery.^{11,13,30,52,60,61,86} Timing of treatment and the specific management of an acute surgical condition in a pregnant patient should be made by a multidisciplinary team, involving surgeons, obstetricians, neonatologists and anaesthesiologists.^{13,25,30,66,86}

Traditionally, the second trimester is considered as the safest period to perform surgery: older studies^{116,134} reported an increased risk of spontaneous abortion in the first trimester and the risk of preterm labour was found to be higher in the third trimester.^{3,8,11,13,22,23,25,26,30,52,60,61,86} In the third trimester, abdominal surgery also becomes more challenging because of the enlarged uterus.^{25,30,52}

Whenever possible, locoregional anaesthesia should be considered, because it avoids the risks of difficult or failed intubation, aspiration, awareness and potential anaesthetic-induced neurotoxicity.^{7,22–26,30,35,52} Furthermore, it provides excellent postoperative analgesia.^{21,23–25,52} If general anaesthesia cannot be avoided, textbooks still advise to perform a classic rapid sequence induction. However, gentle bag-mask ventilation and the use of second-generation laryngeal masks on specific indication (nonobese, starved patients with no other risks for aspiration) are increasingly and safely used in routine clinical practice.^{37–48} Gastric ultrasound can help to support the choice of locoregional anaesthesia or to modify the airway management.³⁶

The risks of both adverse pregnancy outcomes⁴ and anaesthesia-induced neurotoxicity¹¹⁰ are associated with the duration of exposure to anaesthesia. Therefore, all measures need to be taken to keep the duration of exposure as short as possible (e.g. most experienced surgeon, start general anaesthesia only when surgeons are ready to perform and disinfection and draping are completed).

Strengths and limitations

We acknowledge that the present review has several limitations. Although we were able to identify many publications on the anaesthetic management, most reflected expert opinions, most often relying on studies on anaesthesia for caesarean section. The evidence on nonobstetric surgery requiring general anaesthesia remains limited.^{11,13}

Strengths of the narrative review include the database search was carried out according to the standards of a systematic review and the information of all eligible publications of the past 20 years was included to minimise selection bias.

Future studies

The following research priorities have been identified. Firstly, there is no evidence on a specific maternal blood pressure target and how best to achieve this (vasopressors, inotropes, and fluids). Secondly, the comparative risks of sevoflurane versus propofol for preterm labour (uterine relaxation by sevoflurane) and risk of neurotoxicity are not known. Thirdly, there is only limited evidence^{1,135} on the outcome after locoregional anaesthesia versus general anaesthesia. Lastly, only one clinical study investigated the neurodevelopmental outcome after surgery during pregnancy.¹³³

Conclusion

General anaesthesia for nonobstetric surgery during pregnancy is not uncommon. Normal maternal physiology needs to be maintained during general anaesthesia. Several risks are associated with surgery during pregnancy, but it is not known if this is the consequence of the treatment, the disease or exposure to anaesthesia. Postponing urgent surgical interventions during pregnancy is associated with a higher complication rate for both mother and foetus.

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